Hide menu
Introduction
Adapting to Data
Mitigating Training-Serving Skew
Debugging a Production Model
Video: Diagnosing a production model 4 min
Quiz: Designing adaptable ML systems Submitted
Reading: Designing adaptable ML systems 10 min

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Designing adaptable ML systems

coach

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1. Which of the following models are susceptible to a feedback loop? Check all that apply. **1 / 1 point**

- [ ] An election-results model that forecasts the winner of a mayoral race by surveying 2% of voters after the polls have closed.
- [x] A book-recommendation model that suggests novels its users may like based on their popularity (i.e., the number of times the books have been purchased).

Correct
Correct! Book recommendations are likely to drive purchases, and these additional sales will be fed back into the model as input, making it more likely to recommend these same books in the future.

- [x] A university-ranking model that rates schools in part by their selectivity (the percentage of students who applied that were admitted).

Correct
Correct! The model's rankings may drive additional interest to top-rated schools, increasing the number of applications they receive. If these schools continue to admit the same number of students, selectivity will increase (the percentage of students admitted will go down). This will boost these schools' rankings, which will further increase prospective student interest, and so on…

- [ ] A face-attributes model that detects whether a person is smiling in a photo, which is regularly trained on a database of stock photography that is automatically updated monthly.
- [x] A traffic-forecasting model that predicts congestion at highway exits near the beach, using beach crowd size as one of its features.

Correct
Correct! Some beachgoers are likely to base their plans on the traffic forecast. If there is a large beach crowd and traffic is forecast to be heavy, many people may make alternative plans. This may depress beach turnout, resulting in a lighter traffic forecast, which then may increase attendance, and the cycle repeats.

- [ ] A housing-value model that predicts house prices, using size (area in square meters), number of bedrooms, and geographic location as features.

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2. Suppose you are building an ML-based system to predict the likelihood that a customer will leave a positive review. The user interface that customers leave reviews on changed a few months ago, but you don't know about this. Which of these is a potential consequence of mismanaging this data dependency? **1 / 1 point**

- ( ) Change in ability of model to be part of a streaming ingest
- ( ) Change in model serving signature
- (•) Losses in prediction quality

Correct
Correct! For example, a review might be easier to write, and so your prediction of whether someone will leave a review (whether good or bad) is too low because it was trained on reviews that resulted from the older, harder-to-use user interface

3. What is training skew caused by? **1 / 1 point**

- ( ) The prediction environment is slower than the training environment.
- (•) Your development and production environments are different, or different code is used in the training environment than in the development environment.
- ( ) The Cloud Storage you load your data from in the training environment is physically closer than the Cloud Storage you load your data from in the production environment.
- ( ) Starting and stopping of the processing when training the model.

Correct
Correct! Different versions may cause predictions to be significantly slower or consume more memory in the training environment than in the development environment. Different code may result in different performance.

4. Gradual drift is used for which of the following? **1 / 1 point**

- ( ) A new concept that occurs within a short time
- ( ) A new concept that rapidly replaces an old one over a short period of time
- (•) An old concept that incrementally changes to a new concept over a period of time
- ( ) An old concept that may reoccur after some time

Correct
Correct!

5. Which of the following tools help software users manage dependency issues? **1 / 1 point**

- ( ) Modular programs
- (•) Maven, Gradle, and Pip
- ( ) Monolithic programs
- ( ) Polylithic programs

Correct
Correct!

6. Which component identifies anomalies in training and serving data and can automatically create a schema by examining the data? **1 / 1 point**

- ( ) Data ingestion
- ( ) Data transform
- (•) Data validation
- ( ) Data identifier

Correct
Correct!

7. What is the shift in the actual relationship between the model inputs and the output called? **1 / 1 point**

- ( ) Data drift
- (•) Concept drift
- ( ) Prediction drift
- ( ) Label drift

Correct
Correct!